Process Discovery

Lecture 05

BUSINESS PROCESS MANAGEMENT

IS 2206

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Recap: What We Learnt So Far..

Importance of the BPM course

What is the business process?

Background knowledge on the BPM (Modern BPM trends, BPM Suits, As-is and To-be Process Models)

A summarized description of the stages in the BPM Life Cycle

Lecture 1 &
Lecture 2

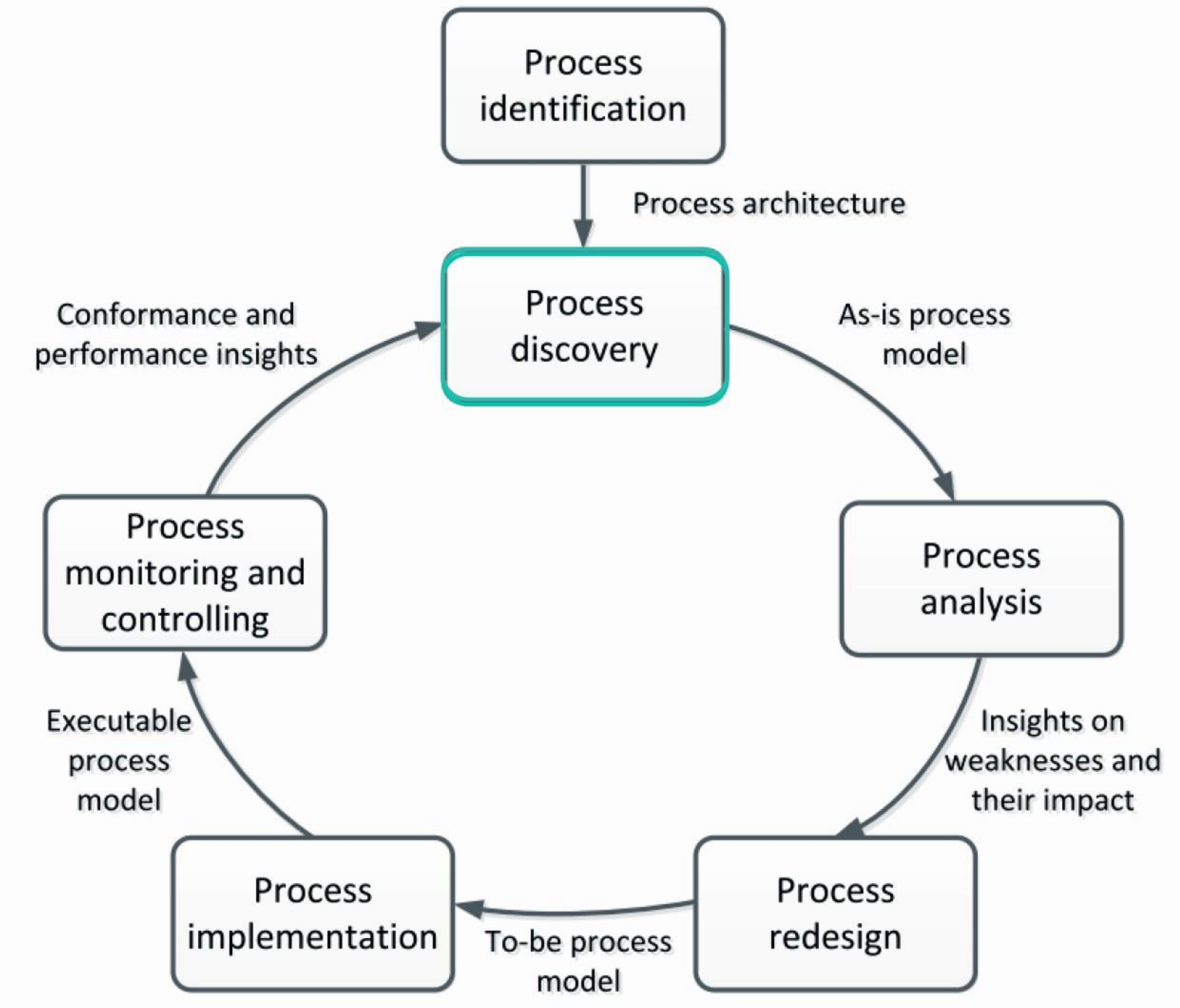
Process Identification

- Types of business processes
- Process architecture in three levels
- Dijckman approach to creating the process architecture in level 1

Lecture 3

Business Process modelling (Business Process Model and Notation)

Lecture 4


Process identification
Process architecture
Process discovery
As-is process model
Process analysis
Insights on weaknesses and their impact
Process redesign
To-be process model
Process implementation
Executable process model
Process monitoring and controlling
Conformance and performance insights

OUTLINE

1. What is Process Discovery
2. Process Discovery Phases
3. Process Discovery Challenges
4. 7 Process Modelling Guidelines - 7PMG

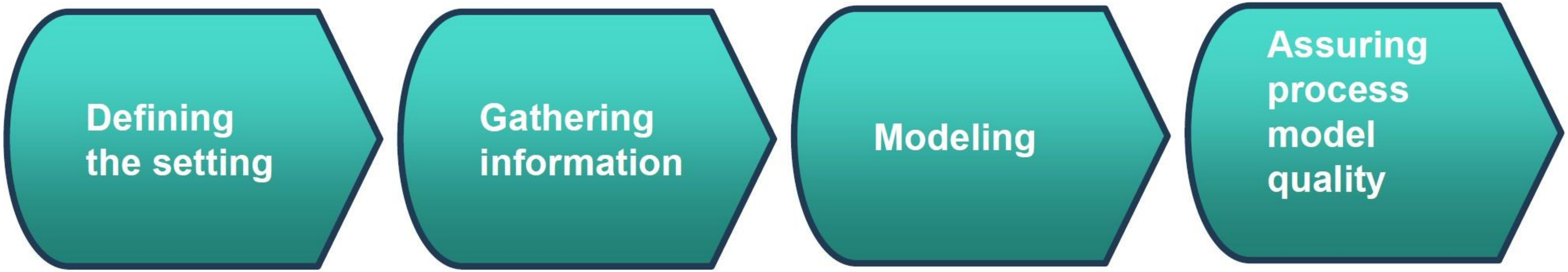
TOPIC 01

What is Process Discovery

1.1 What is Process Discovery?

- The act of gathering information about an existing process and organizing it in terms of an as-is process model.
- The process discovery phase of the BPM lifecycle identifies how to create process models that are both **correct and complete**.
- Modeling is a part of the process discovery.
- Process discovery in general belongs to the category of ill-defined problems. This means in the beginning of a process discovery project, it is not exactly clear
  - Who of the domain experts have to be contacted,
  - Which documentation can be utilized, and
  - Which agenda the different stakeholders might have in mind.

1.2 Introduction to Stages in Process Discovery



Assembling a team in a company.

Building an understanding of the processes through different discovery methods

Organizing the creation of process model. Mapping out the process in a systematic way.

Guarantee that resulting process models meet different quality criteria. Establishing trust in process model

1.2 Introduction to Stages in Process Discovery (Continued)

1. **Defining the setting:** assembling members into a team in a company that will be responsible for gathering information on the business processes.
2. **Gathering information:** This stage is concerned with building an understanding of the process. Different discovery techniques can be used to acquire information on a process.
3. **Conducting the modeling task:** This stage deals with organizing the creation of the process model based on the gathered information . The modeling method gives guidance for mapping out the process in a systematic way.
4. **Assuring process model quality:** This phase aims to guarantee the quality of the process models are created. This phase is important for establishing trustworthiness in the process model.

TOPIC 02

Process Discovery Phases

2.1 Defining the Setting : Appoint a Team

- The team should have members with the knowledge on both business processes and process modeling.

Team = Members who know the processes well + Members who have modeling knowledge well

2.1 Defining the Setting : Appoint a Team

Who are Involved?

Domain Expert

The Domain Expert knows how the business actually works.

ProcessAnalyst

The Process Analyst knows how to model processes.

Neither can succeed without the other.

2.1 Defining the Setting : Appoint a Team
Who are Involved – Process Analyst

1. A Process Analyst is like a translator.
2. The analyst listens to business experts and converts their knowledge into process diagrams.
3. A process analyst is familiar with languages like BPMN and skilled in organizing information in terms of a process diagram.
4. However, process analysts have typically a limited understanding of the concrete process that is supposed to be modeled.
5. For this reason, they depend upon the information being provided by domain experts when creating the process models.

2.1 Defining the Setting : Appoint a Team
Who are Involved – Domain Experts

1. Domain experts have detailed knowledge of the operation of the considered business process.
2. They have a clear understanding of what happens within the boundaries of the process, which participants are involved, which input is required, and which output is generated.
3. On the downside, the domain expert is typically not familiar with languages like BPMN.
4. In some companies, domain experts even refuse to discuss process models and diagrams.
5. Because they feel more comfortable by sticking to natural language for explaining what is happening in the process.

2.1 Defining the Setting : Appoint a Team

Who are Involved – Process Analyst Vs Domain Experts

Aspect	Process Analyst	Domain Expert
Modeling Skills	Strong	Limited
Process Knowledge	Limited	Strong

2.2 Gather Information – Evidence Based Discovery: Document Analysis

- Documents point to existing roles, activities and business objects.
- Formal documentation in terms of
  - Organization chart
  - Employment plan
  - Quality certificate report
  - Internal policies
  - Glossaries and handbooks
  - Forms
  - Work instructions
- May not be process-oriented and trustworthy.
- Could be used to gather information before approaching domain experts.

2.2 Gather Information – Evidence Based Discovery: Document Analysis (Continued)

Advantages	Disadvantages
Highly Available	Some documents are outdated
Useful to gain historical data	Lack of reliability
Low cost	Some documents are not process-oriented
The team can use document when needed again and again	

2.2 Gather Information – Evidence Based Discovery: Observation

- Observe what people do at their workplace
- The team triggers the execution of a process and records the steps that are executed and the set of choices that are offered.
- Trace business objects in the course of their lifecycle.
- Inspect the work environment.
- Observation is particularly useful because it reveals what people actually do rather than what they think they do.
- It allows analysts to gain a realistic understanding of process execution and identify activities that may not be documented elsewhere.

2.2 Gather Information – Evidence Based Discovery: Observation (Continued)

Advantages	Disadvantages
Gather information in actual working environment	People pretends under the observation
Context-rich insight into process	Potentially intrusive

2.2 Gather Information – Evidence Based Discovery: Automated Discovery via Process Mining

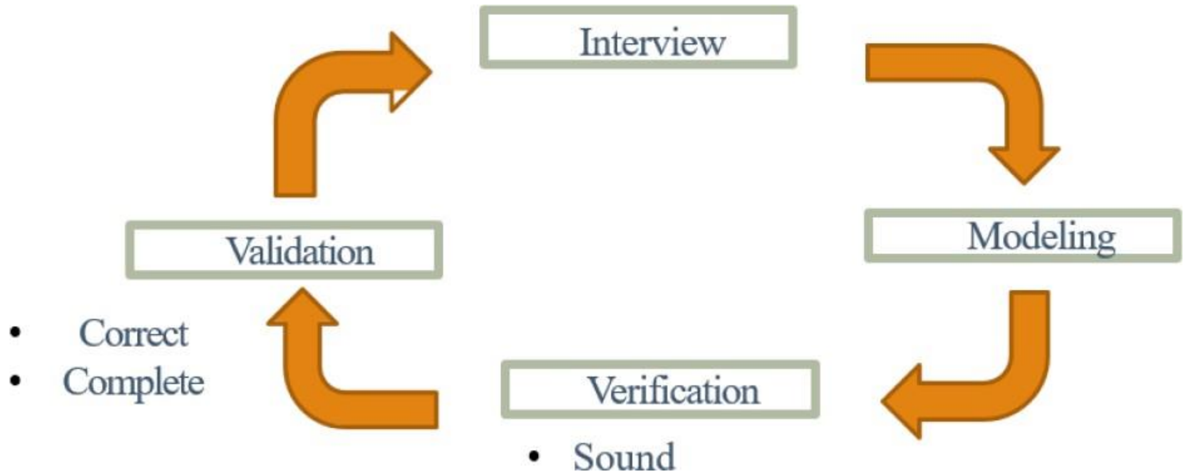
- The basic idea is to extract knowledge from event logs recorded by an information system
- Process mining aims at improving this by providing techniques and tools for discovering process, control, data, organizational, and social structures from event logs
- Automatic process discovery makes use of event logs that are stored by information systems
- Such data must be recorded in a way that each event can be exactly related to an individual case of a project, the specific activity of process and precise point in time

2.2 Gather Information – Evidence Based Discovery:
Automated Discovery via Process Mining (Conutnued)

Advantages	Disadvantages
Event logs capture the execution of a process very accurately	Some event logs are summarized with the term noise
This method provides more accurate information	Potential issue with data quality (Ex: When system crashes , logs are not stored correctly)

2.2 Gather Information – Interviews

- Interviewing domain experts about how a process is executed.
- Structured vs. unstructured interviews
- Assumption: analyst and stakeholder share terminology
- Questions target at identifying deviations from standard processing
- The analyst asks questions to identify activities, participants, decisions, exceptions, inputs, outputs, and business rules associated with the process.
- Effective interviewing requires careful preparation and active listening



2.2 Gather Information – Interviews (Continued)

Advantages	Disadvantages
Gather rich information	Requires sparse time of process stakeholders
The team can identify facial expressions , behavior and body movements of the interviewee.	Time-consuming to conduct interviews
Talking to domain experts offers a view into the history of the process and the surrounding organization	Several iterations required before sign-off

2.2 Gather Information – Workshops

- Gather all key stakeholders together.
- The ***facilitator*** takes care of organizing the verbal contributions of the participants. The ***tool operator*** is responsible for directly entering the discussion results in the modelling tool. Several ***domain experts*** also participate.
- One technique to engage the workshop participants in the discovery effort is by asking workshop participants to build a map of the process using sticky notes.
- The facilitator starts with a pad of sticky notes. Each sticky note is meant to represent a task or event. The group starts to discuss how the process typically starts. The participants start mentioning one or more possible tasks one by one to build the process models.
- A model is directly created during the workshop using any BPM tool.• Model is used as a reference point for discussions.

2.2 Gather Information – Workshops (Continued)

Advantages	Disadvantages
Gather a rich set of information	Conflict of opinion and made by different workshop participants
Collect different opinions from the paricipants	Time-consuming to conduct workshops.

Activity

- Identify strength and weaknesses for each technique.
  - ✓ Document Analysis
  - ✓ Observation
  - ✓ Automated Discovery
  - ✓ Interviews
  - ✓ Workshops

2.2 Gather Information - Summary

Technique	Strength	Weakness
Document Analysis	• Independent from availability of stakeholders • Unbiased information	• Outdated material • Wrong level of abstraction • Less reliable
Observation	• Context-rich insight into process	• Stakeholders likely to behave differently
Automatic Discovery	• Extensive set of cases • Objective data	• Potential issue with data quality
Interview	• Detailed inquiry into process	• Requires sparse time of process stakeholders • Several iterations required before sign-off
Workshop	• Direct resolution of conflicting views	• Synchronous availability of several stakeholders

2.2 Gather Information - Summary

Aspect	Evidence	Interview	Workshop
Objectivity	high	medium-high	medium-high
Richness	medium	high	high
Time Consumption	low-medium	medium	medium
Immediacy of Feedback	low	high	high

2.2 Gather Information - Summary

- It is important to recognize that no single technique can provide a complete understanding of a business process.
- By combining multiple discovery approaches, analysts can cross-check information and improve confidence in the resulting process model.
- The objective is to gather sufficient information to accurately represent the process.

2.3 Modelling

- After gathering information, analysts begin constructing the process model.
- The modeling method provides a structured approach for transforming collected information into a formal process representation.

2.3 Modelling - Stages of Process Modeling Method

1. Identify the process boundaries.
2. Identify activities and events.
3. Identify resources and their handovers.
4. Identify the control flow.
5. Identify additional elements.

2.3 Modelling - Stages of Process Modeling Method

Stage 1: Identify Process Boundaries

- Identifying the process boundaries is essential to identifying the process scope.
- What data are required as input and output to the process?
- Identify the events that trigger our processes.
- Identify the possible process outcomes.
- Identify the starting point and ending point of the process.

2.3 Modelling - Stages of Process Modeling Method

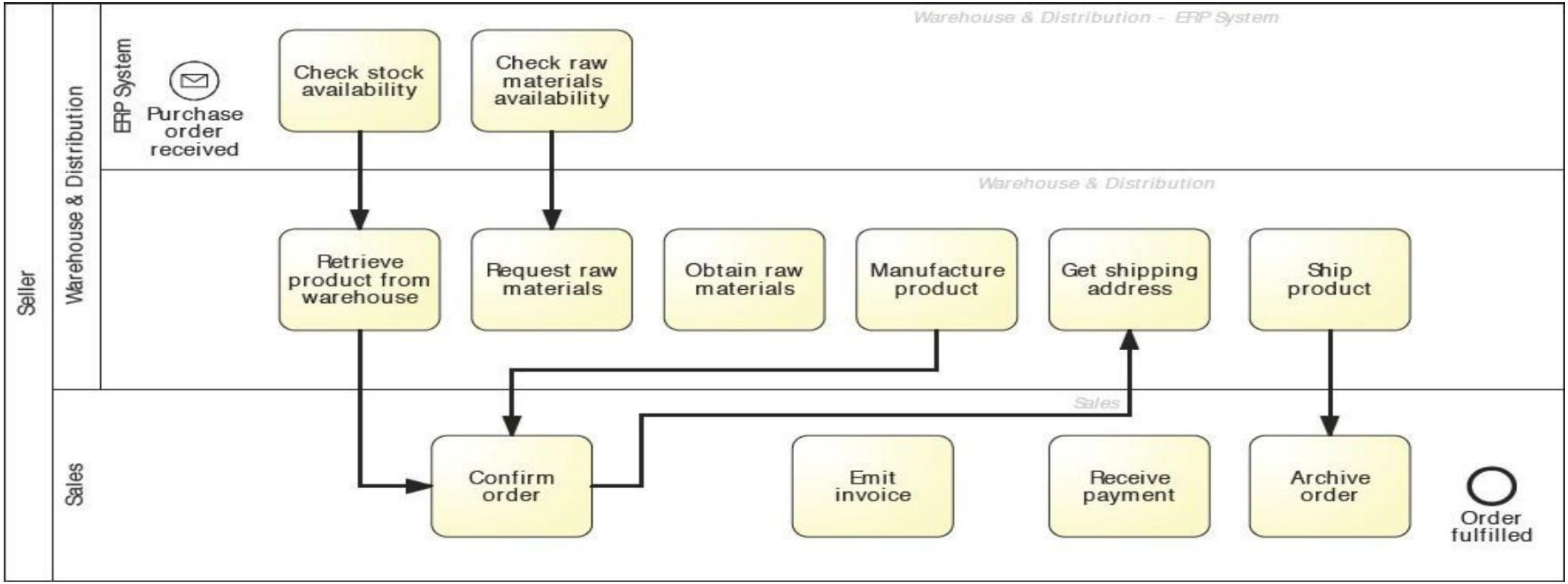
Stage 2: Identify Activities and Events

- At this stage, analysts focus on identifying all major tasks and important events that take place throughout the process.

2.3 Modelling - Stages of Process Modeling Method

Stage 3: Identify Resources and their handovers

- Identify who is responsible for each and every task in the business process



2.3 Modelling - Stages of Process Modeling Method

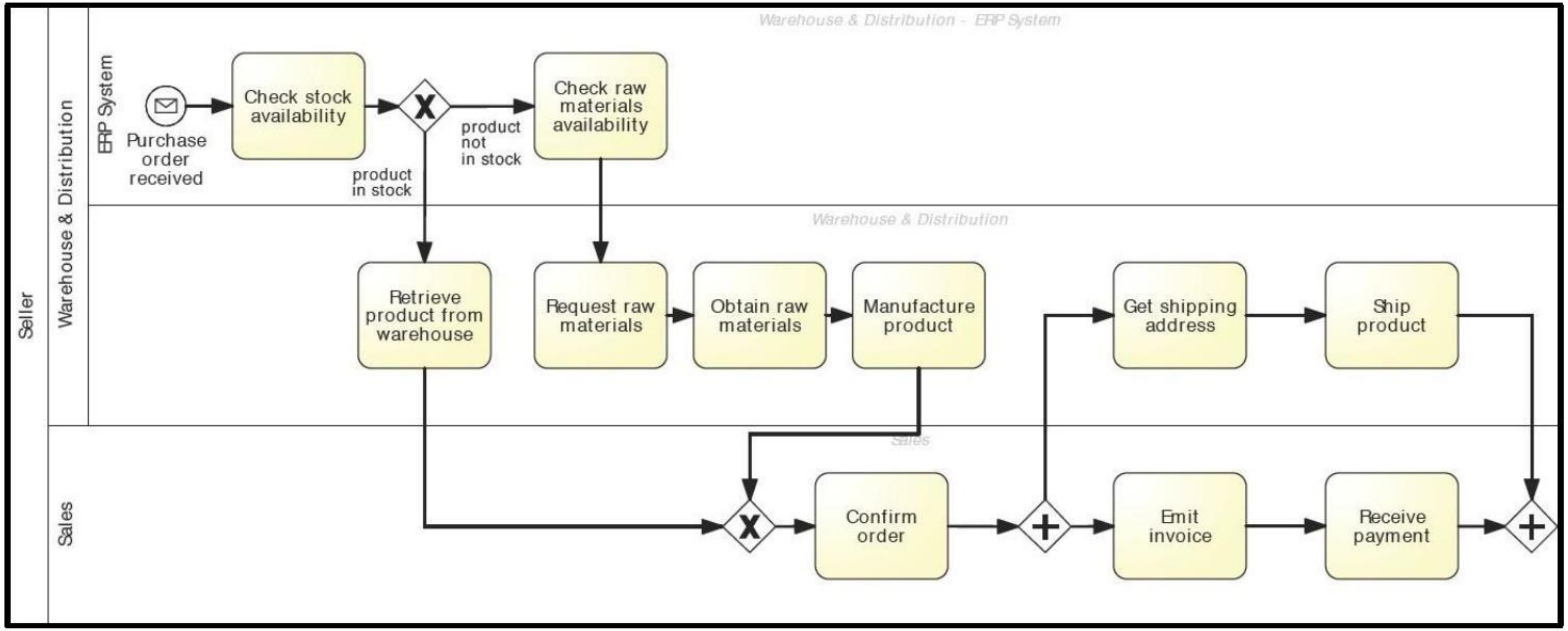
Stage 3: Identify Resources and their handovers

- Resources refer to the people, roles, departments, or systems responsible for performing activities.
- Handovers occur when responsibility moves from one participant to another.
- Understanding resources and handovers helps analysts represent organizational responsibilities accurately.

2.3 Modelling - Stages of Process Modeling Method

Stage 4: Identify the Control Flow

- When and Why activities are executed



2.3 Modelling - Stages of Process Modeling Method

Stage 4: Identify the Control Flow

- Control flow describes the sequence in which activities are performed.
- At this stage, analysts determine the logical relationships between activities.
- They identify the order of execution, decision points, parallel paths, and alternative flows.
- The control flow forms the backbone of the process model.

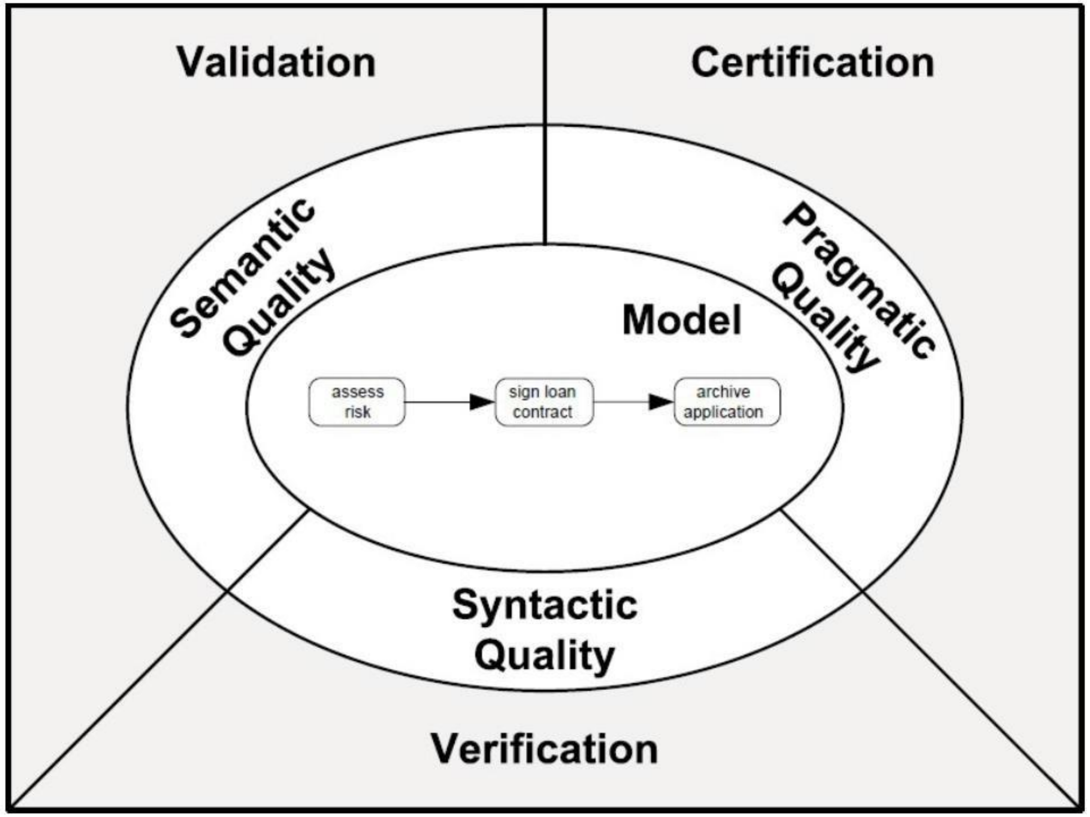
2.3 Modelling - Stages of Process Modeling Method

Stage 5: Identify Additional Elements

- Extend the process model by capturing artifacts and exception handlers.
- Artifacts – data objects, groups, text annotations and their relations with other elements in the process model.
- Exception Handlers – boundary events, exception flows.

2.4 Ensure Quality of the Process Models

- Gathering information and organizing it in a process model is often done in a sequential way.
- There is a need for various steps of quality assurance.
- There are three types of qualities: Syntactic quality, semantic quality, pragmatic quality.



2.4 Ensure Quality of the Process Models

Syntactic Quality: Verification

- Syntactic quality relates to producing a process model that confirms to syntactic rules and guidelines defined by the process modeling language.

  For Example : In BPMN it is not allowed to draw a sequence flow across the boundaries of pools. BPMN defines an extensive set of syntax rules.

- Verification essentially addresses formal properties of a model that can be checked without knowing the real-world process.
- But Verification checks the behavioural and structural correctness of the process.
  - **Structural correctness** relates to the types of elements that are used in the model and how they are connected.
  - **Behavioural correctness** relates to potential sequences of execution as defined by the process model.

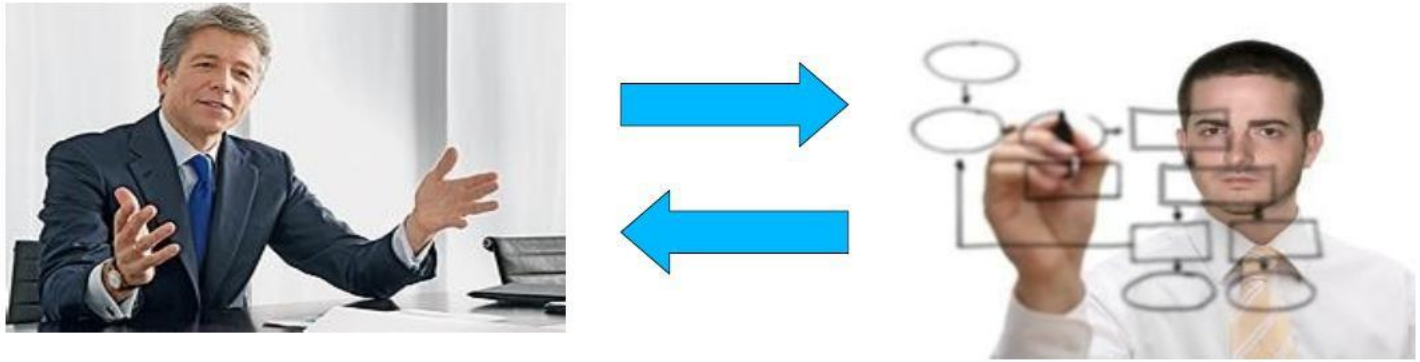
2.4 Ensure Quality of the Process Models

Semantic Quality: Validation

- Semantic quality relates to the goal of producing models that make true statements about the considered domain, either for existing as-is processes or future to-be processes.
- The process model has to be compared with the real-world domain of a particular business process.
- There is no set of formal rules that can be used to easily check semantic quality.

Validity

Completeness



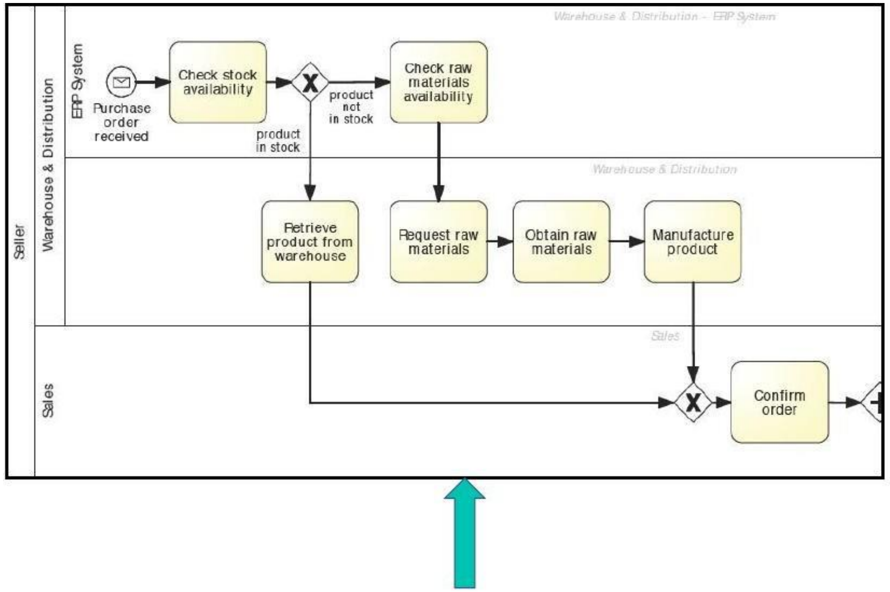
Domain Expert

Process Analyst

2.4 Ensure Quality of the Process Models

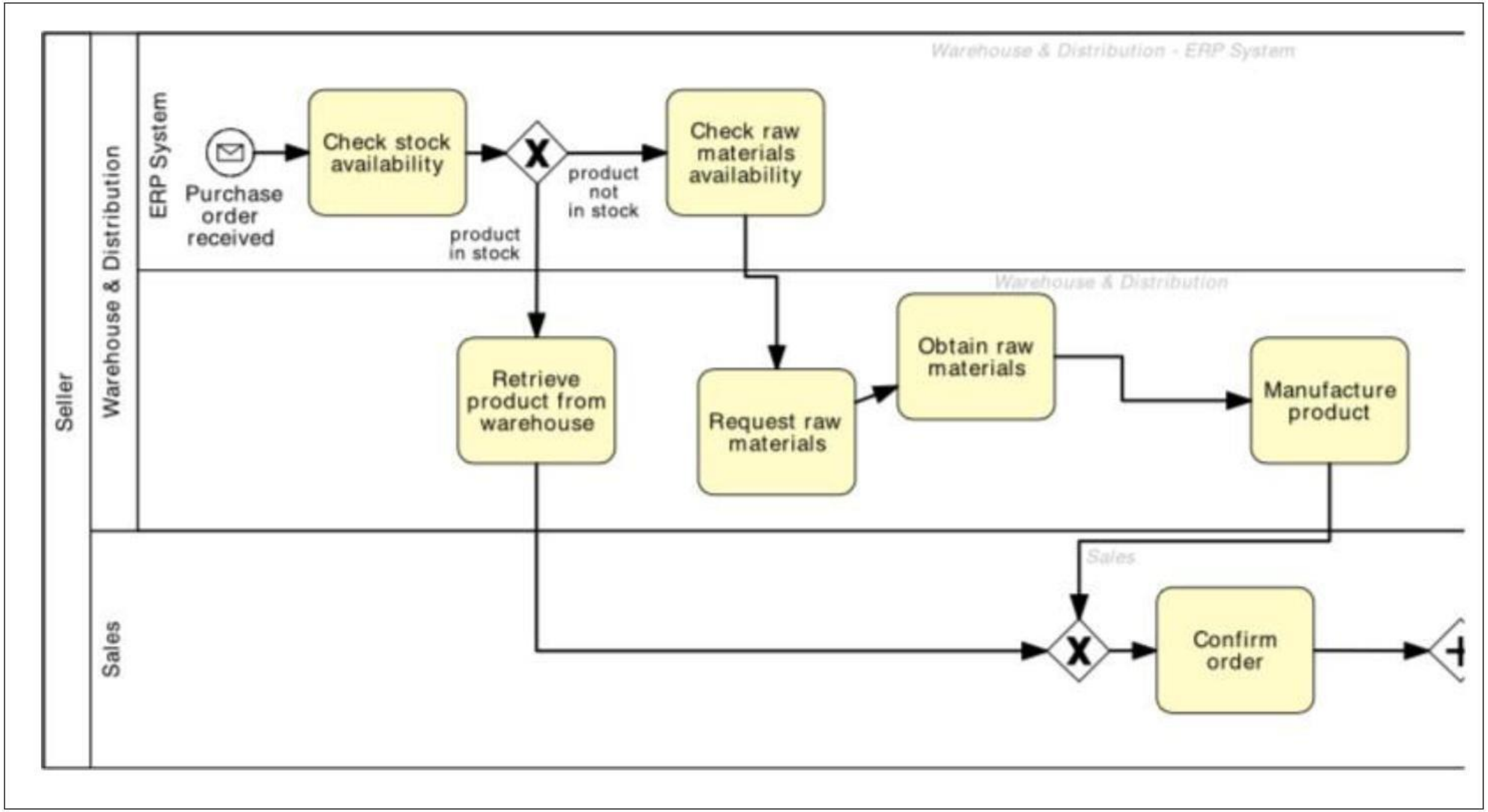
Pragmatic Quality: Layout

- Pragmatic quality refers to the goal of building a process model with good usability.
- Certification is the process of checking the pragmatic quality of a process model by investigating its usage.
- Usability including understandability, maintainability, and learning.
- Understandability relates to the fact of how easy it is to read a specific process model.
- Maintainability points to the ease of applying changes to a process model.
- Learning relates to the degree of how good a process model reveals; how a business process works in reality.

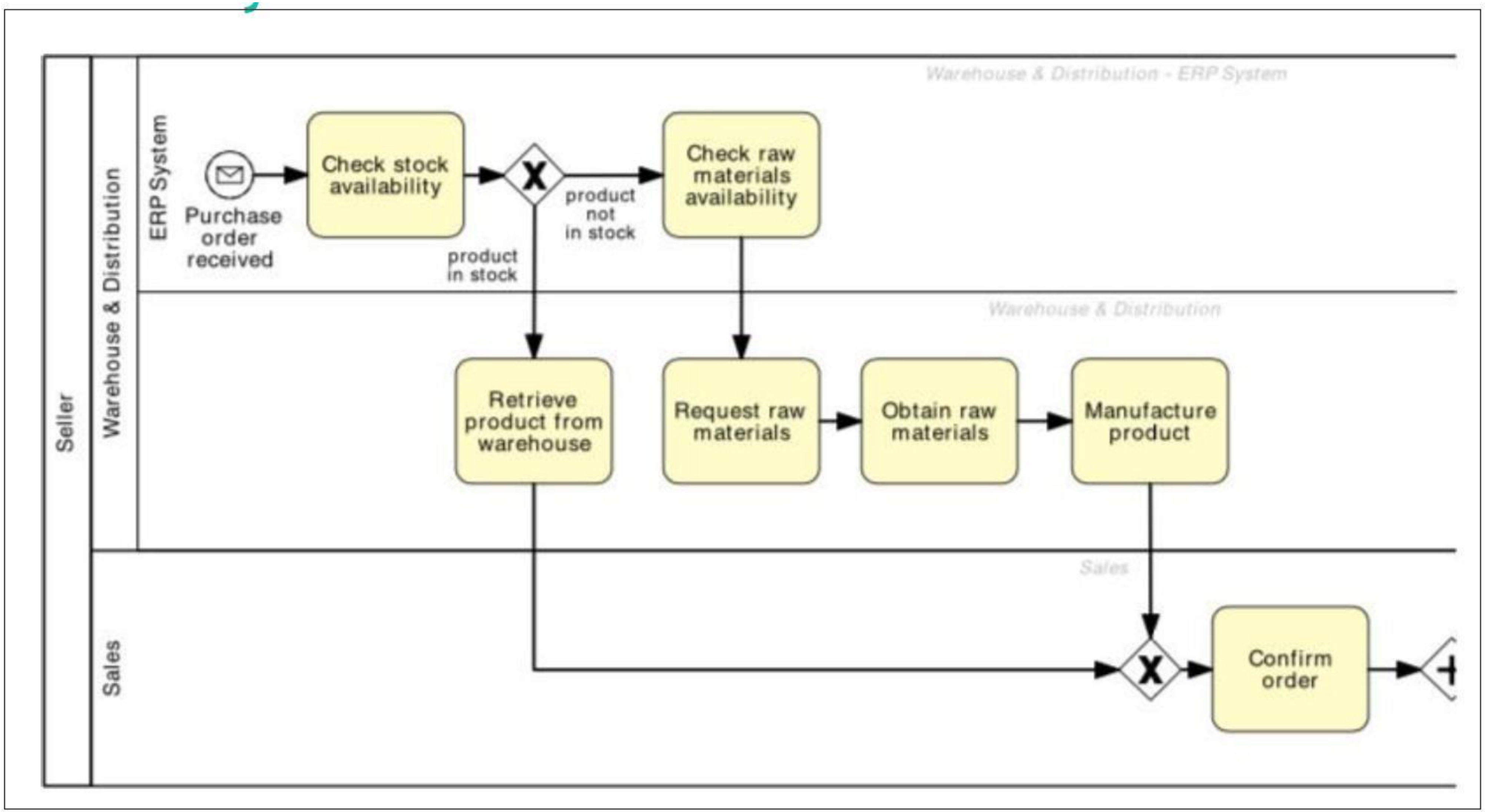


Models must look nice

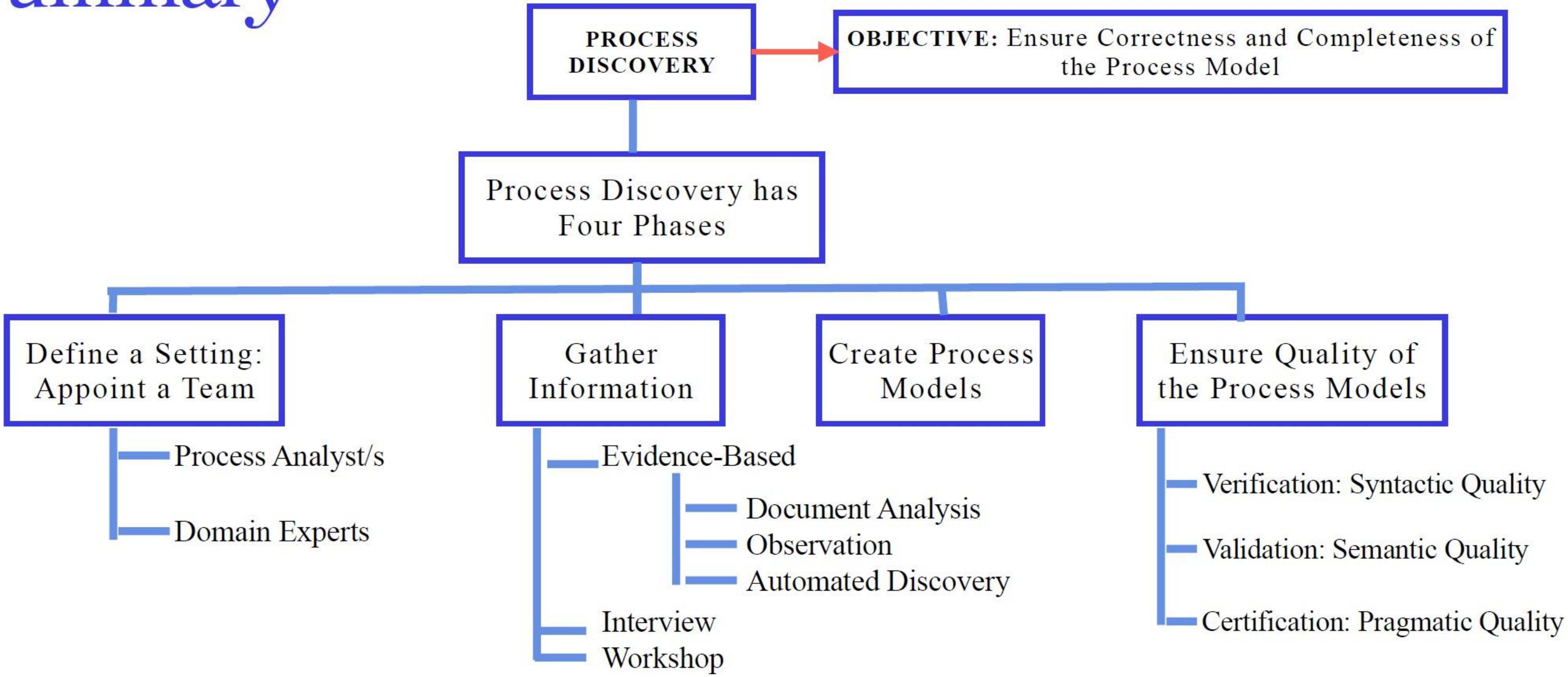
Example: Bad Layout



Example: Correct Layout



Summary



PROCESS DISCOVERY
OBJECTIVE: Ensure Correctness and Completeness of the Process Model
Process Discovery has Four Phases
Define a Setting: Appoint a Team
Gather Information
Create Process Models
Ensure Quality of the Process Models
Process Analyst/s
Domain Experts
Evidence-Based
Document Analysis
Observation
Automated Discovery
Interview
Workshop
Verification: Syntactic Quality
Validation: Semantic Quality
Certification: Pragmatic Quality

TOPIC 03

Process Discovery Challenges

Process Discovery Challenges

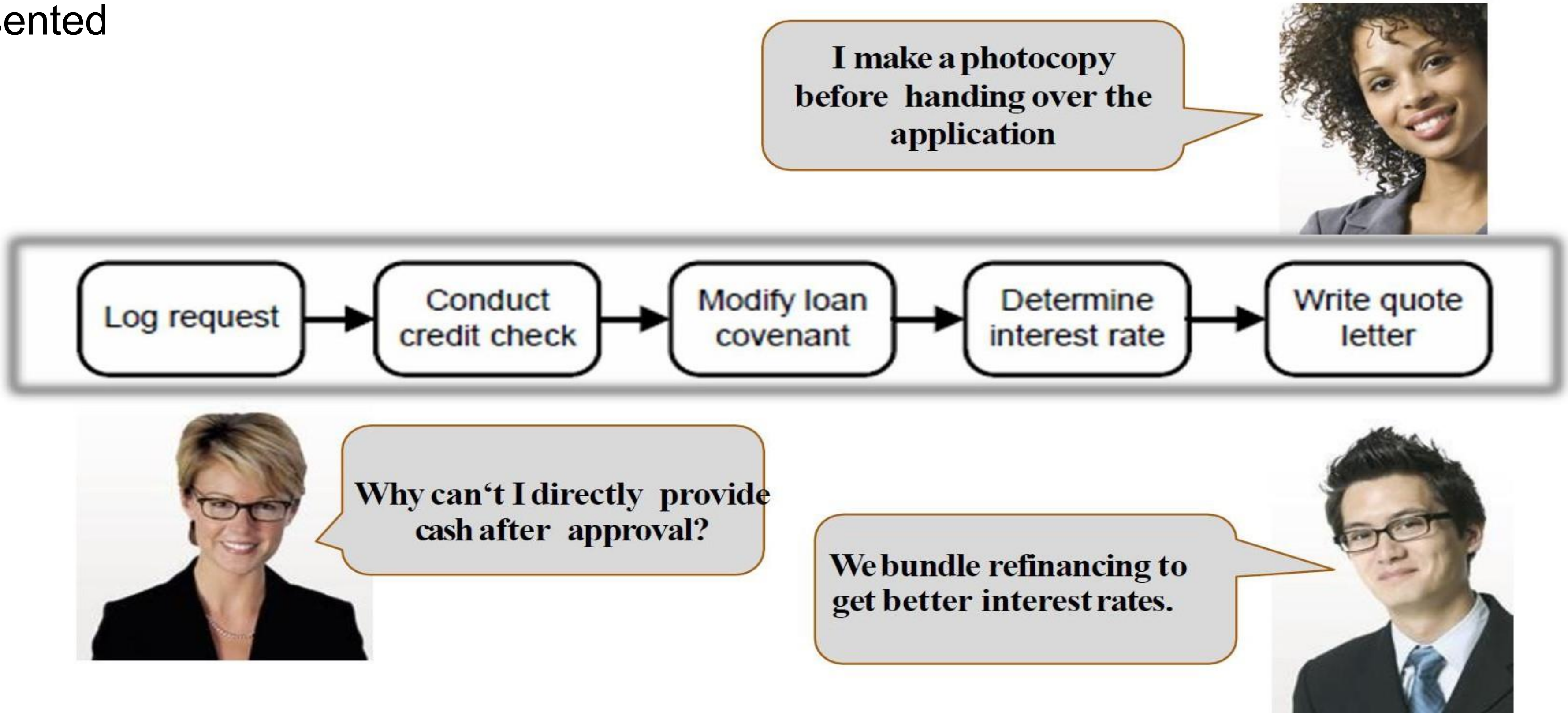
1. Fragmented process knowledge.
2. Domain experts think on case level.
3. Knowledge about process modeling is rare.

3.1 Challenge 1: Fragmented Process Knowledge

- A business process defines a set of logically related activities.
- These activities are assigned to specialized participants.
- A process analyst needs to gather information about a process not only by talking with a single domain expert but with several domain experts who are responsible for the different tasks of the process.
- Typically, domain experts have an abstract understanding of the overall process and a very detailed understanding of their own task.

3.1 Challenge 1: Fragmented Process Knowledge (Continued)

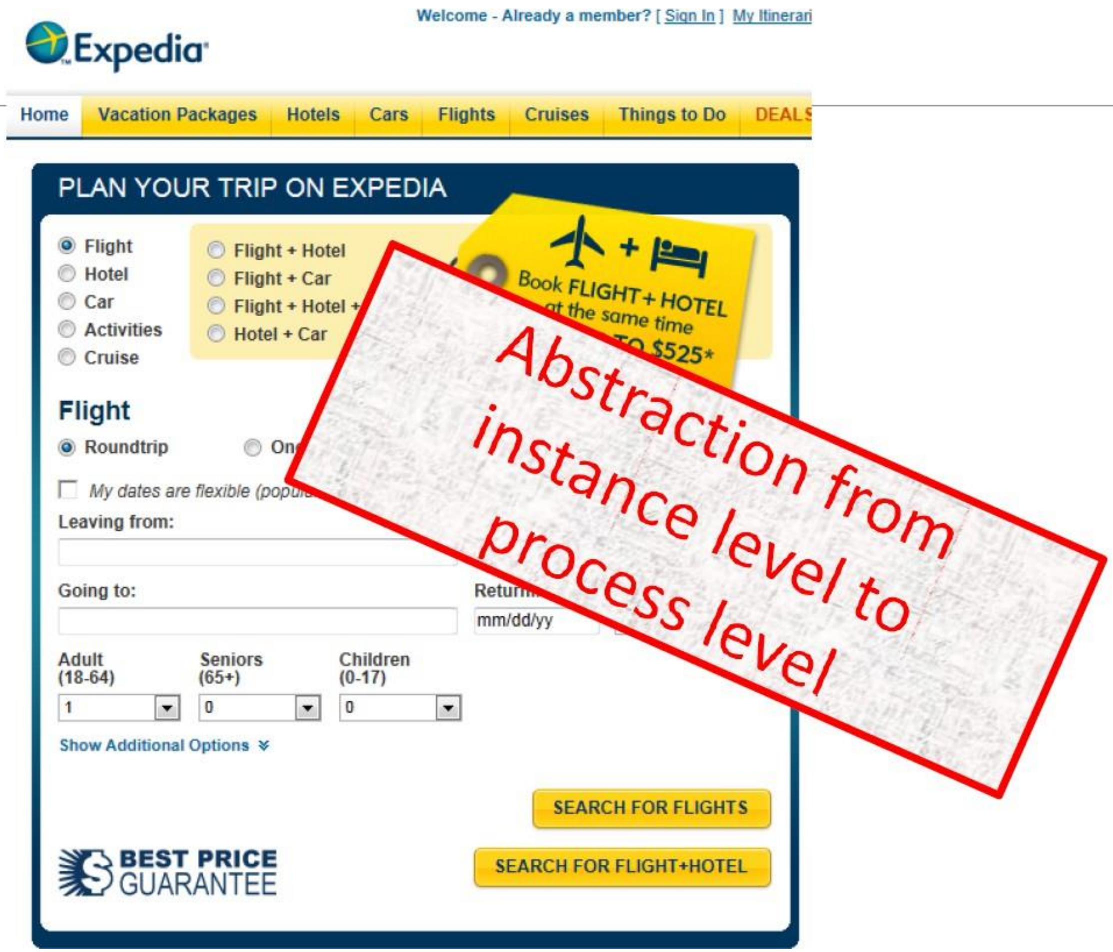
- This fragmentation of knowledge requires analysts to combine different perspectives into a single coherent process model.
- The challenge lies in ensuring that all relevant activities, interactions, and dependencies are accurately represented



3.2 Challenge 2: Domain Experts think on case Level

- Domain experts will easily describe the activities of a specific case but they might have problems responding to general questions.
- Process analysts often get answers like “you cannot really generalize, every case is different”.
- Process analyst will have to organize and abstract from the pieces of information provided by the domain experts.

3.2 Challenge 2: Domain Experts think on case Level (Continued)



”Every trip is different.“

”You cannot really compare. Our customers go to different placesin different seasons using different modes of transportation.“

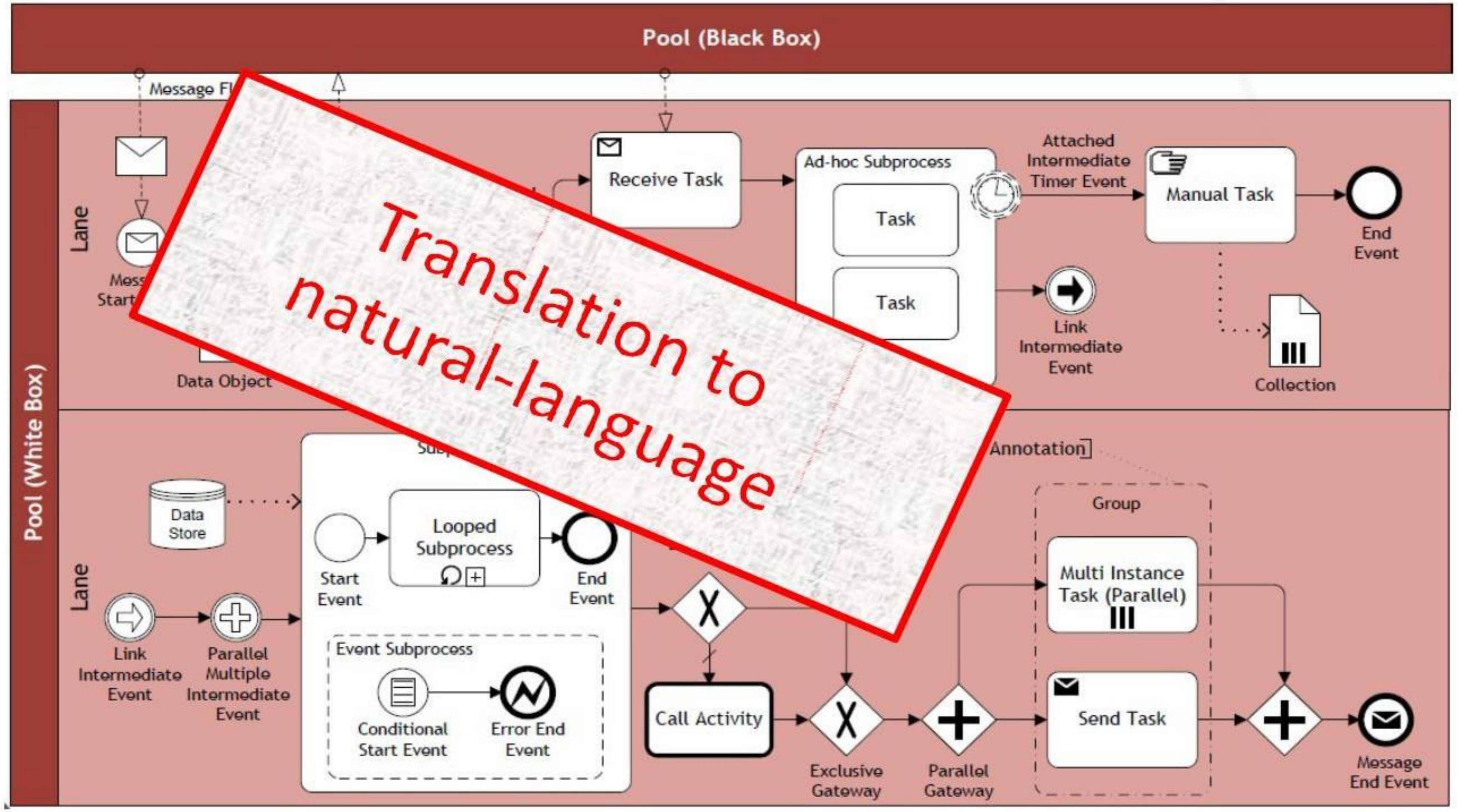
”We can never do anything exactly in the same way. There are so many special conditions.“

3.3 Challenge 3: Knowledge about Process Modeling is rare

- Domain experts are typically not familiar with business process modeling languages.
- Domain experts are often trained to create or read process models.
- Thus seeking feedback to a draft process model is difficult.

3.3 Challenge 3: Knowledge about Process Modeling is rare (Continued)

”Could you please tell me, whether this diagram correctly shows your process?“



TOPIC 04

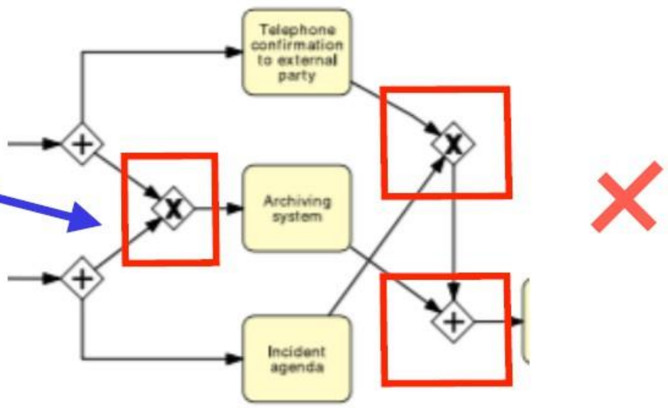
7 Process Modelling Guidelines - 7PMG

7 Process Modelling Guidelines (7PMG)

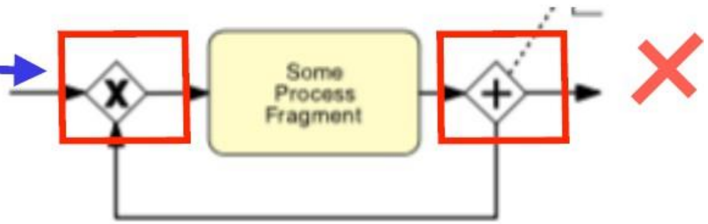
- The purpose of the 7PMG guidelines is to improve the quality and readability of process models.
- These guidelines focus on reducing unnecessary complexity, improving clarity, and increasing understandability.
- By following these principles, analysts can create models that communicate process knowledge more effectively and support process improvement initiatives.

7 Process Modelling Guidelines (7PMG)

G1 - Use a few elements in the model as possible.
G2 - Minimize the routing paths per element.



G3 - Use one start and one end event per path.
G4 - Model as structured as possible.
G5 - Avoid OR routing elements (OR gateways).
G6 - Use verb-noun activity labels.
G7 - Decompose a model with more than 30 elements.



Recap...

Thank You!